

Two honest caveats bound this result. First, the effect size is modest in absolute terms: with 3 seeds and 128 test samples the per-condition standard error is roughly 0.02, so the 0.05 gap to the single-modality baselines is resolvable but not dramatic, and 0.48 is statistically indistinguishable from the 0.50 chance floor. Second, the CNN reaches only 0.53 on single-modality data against a 0.75 calibration target, so the network operates far from the Bayes bound and part of the joint-training deficit could in principle reflect generic overfitting with more trainable parameters. What the direction of the result nevertheless establishes is that joint training does not merely fail to combine the two modalities: it undershoots baselines that demonstrably use only one, which is consistent with active corruption of the spectral pathway rather than mere under-use, and is inconsistent with the “spectral features are simply irrelevant” reading (Remark 21). The frozen-vs-joint ordering, which is the theory’s core prescription, holds in all three calibration modes (Table 1: +0.13, +0.04, +0.09); the undershoot below both single-modality baselines appears specifically in the **bayes** mode, where the modalities are matched most cleanly.

Comparison of calibration modes. Table 1 compares the three calibration modes on two criteria: (i) calibration quality, defined as $1 - |a_{\text{spec}} - a_{\text{spat}}|$ where $a_{\text{spec}}, a_{\text{spat}}$ are the achieved single-modality accuracies; and (ii) shortcut strength, defined as $a_{\text{frozen}} - a_{\text{joint}}$.

mode	α, β	spec / spat	joint	frozen	shortcut
bayes	0.82, 18.75	0.75/0.75	0.48	0.61	+0.13
ntk	0.99, 4.91	0.78/0.58	0.53	0.57	+0.04
margin	5.00, 5.00	0.95/0.59	0.60	0.69	+0.09

Table 1: Calibration mode comparison for the equal-information killer test (CNN $D = 256$, 3 seeds). *spec / spat*: achievable single-modality accuracies. *shortcut* := $a_{\text{frozen}} - a_{\text{joint}}$. The **bayes** mode produces both the cleanest modality equalization and the largest shortcut effect.

Conclusion. The **bayes** calibration delivers the qualitative claim: when spectral and spatial signals carry equal information about the label (measured by what each modality alone can provide), end-to-end joint training underperforms both single-modality baselines, while freezing the spectral module recovers information that joint training squanders. Subject to the effect-size caveats above, this is empirical support consistent with the spectral-shortcut mechanism suggested by Theorem 27.

7.5 Robustness

In supplementary experiments we vary the optimizer (Adam vs. SGD with momentum), the noise level (0.05 vs. 0.10), and the spectral input dimension S . The capacity-driven joint-vs-frozen gap and the EGR collapse pattern hold across all variations, with two notable modulations: SGD makes the EGR signal cleaner (Adam normalizes per-parameter and partially masks the natural gradient ratio), and lower noise allows cross-entropy to fully saturate, accelerating the emergence of the shortcut.

[TODO: Phase 5, W15: write the robustness analyses into the supplement once Exp 1.3 is finalized.]

8 Real-Data Study: Geometry, Dynamics, and Scope

The synthetic experiments in Section 7 verify the theorems where their assumptions hold. This section takes the theory to a production-scale spectral-spatial pipeline: per-pixel 4-class tissue classification on QCL infrared hyperspectral images of breast cancer tissue microarrays. It

contains two kinds of evidence with two distinct roles. First, a production-scale *observation* from a companion empirical study [Dziuba et al., 2026]: end-to-end training of the spectral module loses badly to pretraining and freezing it. Second, a *controlled, instrumented study* on the same data and architecture family, designed to test the theorems’ predictions directly — with matched arms, logged gradient dynamics, and the optimizer placed inside and outside the theorems’ regime. The picture that emerges is sharper than simple confirmation: the block-scalar reading of Theorem 14 does *not* manifest at initialization, for a reason we can measure (rank-one input statistics); a direction-wise reading does manifest, strongly; and Theorem 27’s functional prediction holds in its own regime while joint training outside that regime is *worse* than the theorem predicts — the spectral module is not merely starved but actively degraded.

8.1 Pipeline and capacity

The data are 336×336 pixel cores collected by QCL spectroscopy over the 1000–1800 cm^{-1} fingerprint region. Each pixel carries an $S = 942$ dimensional spectrum (after stacking raw absorbance and its first two derivatives). The classification target is per-pixel tissue type. The compositional model is the Block Vision Transformer of O’Leary et al. O’Leary et al. [2026]: a linear per-pixel $f_{\theta} : \mathbb{R}^{942} \rightarrow \mathbb{R}^K$, then 16×16 patch tokenization, a 12-layer / 12-head Vision Transformer of embedding width $M = 192$, a transposed-convolution decoder, and a convolutional segmentation head. Parameter counts obtained directly from the instantiated model are given in Table 2.

Table 2: Module capacities of the BlockViT-v2 pipeline, counted from the instantiated model. K is the spectral bottleneck width; the two rows are the two configurations used in the companion’s production runs.

Configuration	C_f	C_g	C_g/C_f
$K = 64$ (end-to-end baseline)	61,108	18,271,540	299
$K = 128$ (matched to frozen variants)	121,588	21,472,564	177

Channel convention. O’Leary et al.’s original BlockViT operates on the second-derivative spectrum alone (a single channel, $S = 965$ for their prostate FTIR data)—the field-standard preprocessing in infrared histopathology. The companion benchmark, and hence the models analyzed here, instead stacks raw absorbance with its first two derivatives (three channels, $S = 942$). The choice affects only C_f : instantiating the same model on the single-channel field-standard input gives $C_f = 20,916$ and $C_g/C_f \approx 874$. The three-channel convention is therefore the *conservative* choice for this paper’s purposes—under the field standard, the capacity asymmetry we analyze is roughly three times larger.

The capacity ratio is thus roughly two orders of magnitude, and the spatial module’s dominant blocks—the transformer (5.34×10^6) and the transposed convolution (9.44×10^6)—are both $\Theta(M^2)$ in the embedding width, so this architecture falls in the class treated by Theorem 14. Whether the curvature disparity the theorem permits actually manifests at initialization is an empirical question, which we now measure directly.

8.2 Production-scale observation

The companion study trains the full pipeline under 5-fold cross-validation on breast QCL data and reports held-out test macro-F1 for an end-to-end learned linear spectral module against three pretrain-and-freeze variants (its Table 6):

Two remarks on scope. First, this is an *observation that motivates the controlled study below*, not a theorem test: the variants differ in both stability (frozen vs. jointly trained) and informa-

Variant	Test F1 (5-fold mean $\pm$ sd)
End-to-end learned linear (joint)	0.675 ± 0.036
Frozen pretrained spectral (Peak Windows)	0.842 ± 0.121
Frozen pretrained spectral (MLP)	0.896 ± 0.092
Frozen pretrained spectral (Sliding Win.)	0.954 ± 0.020

Table 3: Held-out test F1 on breast QCL tissue cores, as published in the companion study [Dziuba et al., 2026]. End-to-end training of the spectral module underperforms every pretrain-and-freeze variant by 0.17–0.28 F1.

tiveness (pretrained vs. learned from scratch), and the production runs log no gradient dynamics. Second, the theorems do not guarantee that pretrain-and-freeze wins. Theorem 27’s falsifiable prediction is that joint training behaves like *freezing the spectral module at initialization*; pre-training is a separate, empirically supported escape route. The controlled study disentangles exactly these factors.

8.3 Geometry at initialization

We measure the Gauss–Newton blocks of the instantiated pipeline on real training batches at random initialization (Lanczos with full reorthogonalization on matrix-free GGN block operators, validated to 10^{-12} against dense computation; Section 7), sweeping the ViT embedding width $M \in \{48, 96, 192, 384\}$ with 3 seeds $\times$ 4 batches per width.

The block-scalar disparity does not manifest. At production width $M = 192$ the measured curvature ratio is $\mathcal{D}_{\text{curv}} = \lambda_{\max}(G_{\phi\phi})/\lambda_{\max}(G_{\theta\theta}) = 0.46 \pm 0.21$ — the *spectral* block carries the larger top curvature, the opposite of the naive block-scalar expectation. The ratio rises with width (0.33 ± 0.12 at $M=48$ to 0.70 ± 0.19 at $M=384$; on prostate QCL it reaches parity, 1.09 ± 0.52 , at $M=384$) but the disparity regime is nowhere in sight at buildable widths. Both theorem ingredients behave exactly as proved: the spectral cap is width-independent as Theorem 14 requires (log-log slope of $\lambda_{\max}(G_{\theta\theta})$ vs. M : -0.003 breast, -0.012 prostate), and $\lambda_{\max}(G_{\phi\phi})$ grows with M , though sublinearly over this range ($115 \rightarrow 258$ across an $8\times$ width increase). What fails is the *informativeness* of the bound, and the data tell us why.

Diagnosis: rank-one input statistics. The input second-moment matrix Σ_X of real tissue spectra has effective rank 1.07 (breast) and 1.02 (prostate) out of 942: every tissue spectrum shares the same dominant mean shape. The cap $C_\theta \propto \lambda_{\max}(\Sigma_X)$ is therefore enormous — the inequality of Theorem 14 holds but is vacuous at buildable widths, because the data concentrate all their second-moment mass in one direction. This is a measurable regime violation, not a counterexample: the theorem’s disparity is a statement about what capacity *permits*, and rank-one inputs hand the spectral block a single direction of huge curvature.

Starvation is direction-wise. Precisely because the spectral curvature is rank-deficient, the block-scalar ratio is the wrong lens. Restricting the GGN to informative directions shows the disparity the scalar hides: the spectral block’s top eigenvector aligns 0.954 with the mean-spectrum direction, and the top 40 of 61,108 spectral directions hold 83% of the trace. Along the clinically decisive CancerEpi–CAS class-contrast direction, the spectral curvature is $12\text{--}28\times$ *smaller* than along the mean-spectrum direction; the spatial block’s top eigenvalue overtakes the spectral spectrum already at rank 2–5 (Figure 1). The gradient signal available for learning discriminative spectral features — as opposed to re-encoding the mean spectrum — is starved in exactly the directions that matter.

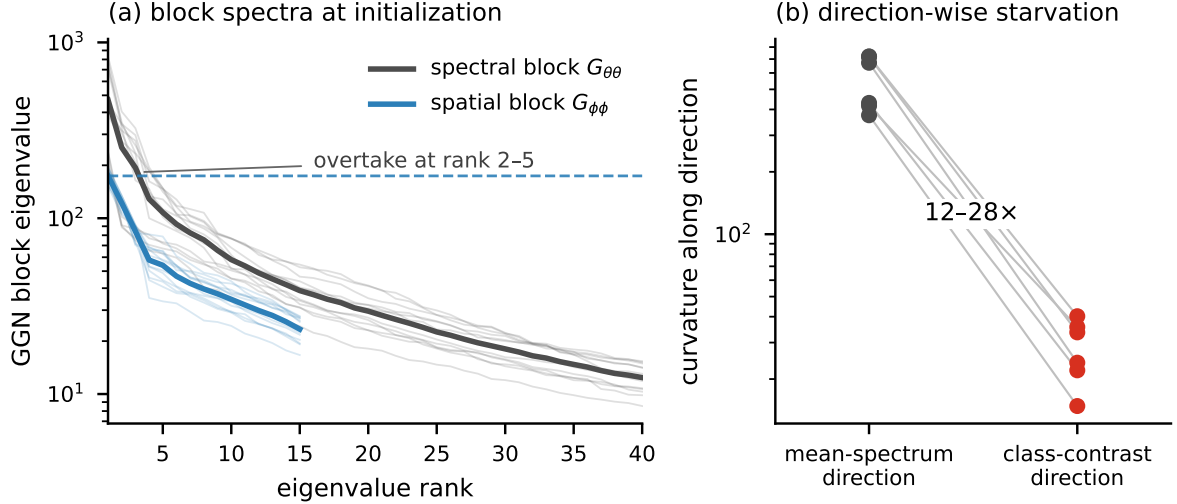


Figure 1: Direction-wise starvation at initialization (breast QCL, $M=192$, 3 seeds $\times$ 4 batches). (a) GGN block eigenvalue spectra: thin lines are individual measurements, thick lines their means; the dashed line marks the spatial block’s top eigenvalue, which overtakes the spectral spectrum at rank 2–5. (b) Spectral curvature along the mean-spectrum direction vs. along the CancerEpi–CAS class-contrast direction, per measurement: the discriminative direction is 12–28 $\times$ starved.

Normalization sets the cap. The cap is not an immutable property of the tissue: removing the spectral normalization deflates $\lambda_{\max}(G_{\theta\theta})$ by $2.4\times$ ($471 \rightarrow 194$ at $M=192$) and pushes the ratio across parity (1.18 at $M=192$, 1.31 at $M=384$). Input *centering*, by contrast, changes nothing when the architecture contains internal batch normalization ($\mathcal{D}_{\text{curv}}$ identical to three decimals with and without centering): a linear projection followed by normalization absorbs constant input shifts exactly. The lever on the curvature geometry acts through the normalization choice, not the preprocessing pipeline.

8.4 Controlled retraining dynamics

We then train the pipeline ourselves on breast QCL (fold 0), with full instrumentation: arms {joint-linear, frozen-random, frozen-PCA, joint-MLP} $\times$ widths $M \in \{48, 192\} \times 3$ seeds under AdamW (60 epochs), plus a {joint-linear, frozen-random} $\times$ 3-seed pair under plain SGD at $M=48$ — the closest buildable match to the theorems’ regime. Following Section 5, the spectral parameters sit in a zero-weight-decay group in every arm; frozen arms log counterfactual spectral gradients. Per-step residuals, block gradient norms, and per-epoch parameter displacements and input-Jacobian norms are recorded.

Table 4: Controlled study: held-out validation macro-F1 (mean $\pm$ sd over 3 seeds, mean of final 5 epochs) and relative spectral-parameter displacement at end of training, AdamW arms.

Arm	$M=48$	$M=192$	$\ \Delta\theta\ /\ \theta_0\ $
Joint (linear)	0.538 ± 0.082	0.646 ± 0.050	0.55–0.60
Frozen random	0.648 ± 0.063	0.731 ± 0.023	0
Frozen PCA	0.731 ± 0.003	0.727 ± 0.012	0
Joint (MLP)	0.566 ± 0.070	0.575 ± 0.184	0.86–0.95

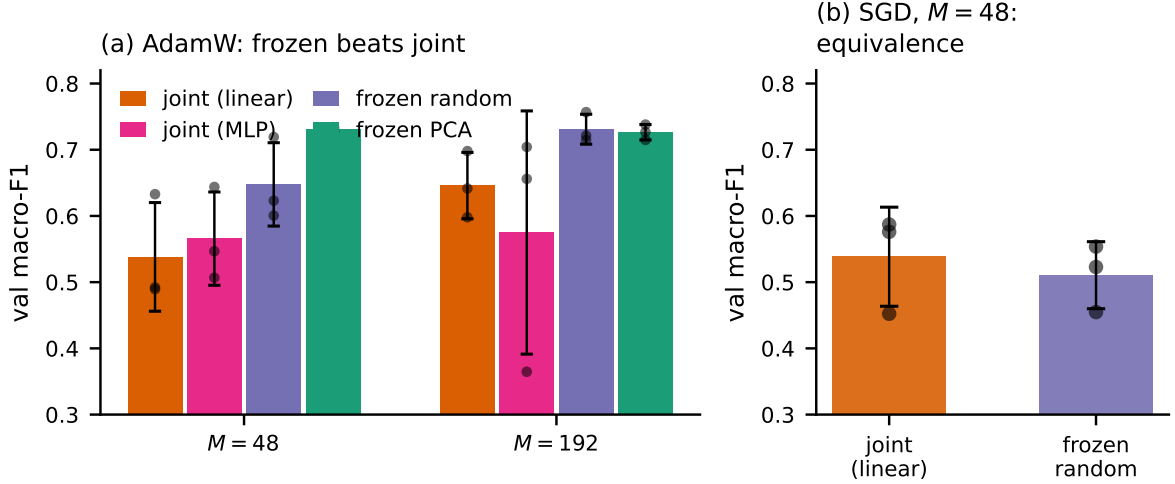


Figure 2: Controlled retraining study (breast QCL fold 0; bars: mean over 3 seeds, whiskers: sd, dots: individual seeds). (a) Under AdamW every frozen arm beats every joint arm at both widths. (b) Under plain SGD with zero weight decay — the regime of Theorem 27 — joint training and frozen-random are statistically indistinguishable.

Frozen beats joint everywhere. Under AdamW, freezing the spectral module at a *random* initialization outperforms training it jointly at both widths (gaps -0.110 at $M=48$, -0.085 at $M=192$; Figure 2a, Table 4). The pre-registered falsifier — joint beating frozen-PCA decisively — comes out negative at both widths (-0.193 , -0.081). The production-scale observation of Section 8.2 survives the controlled setting with informativeness held fixed: *stability alone* accounts for a large fraction of the gap. Indeed at $M=192$ frozen-PCA and frozen-random are statistically indistinguishable (0.727 vs. 0.731); the informativeness of the frozen features matters at small width (0.731 vs. 0.648 at $M=48$) but stability is what the theory speaks to, and stability is what does the work here.

Harmful drift, not lazy pinning. The joint arms’ spectral parameters are not starved into stillness: they move 55–60% in relative norm under AdamW (86–95% for the MLP spectral module) — and this movement *hurts*, since the same module left at its random initialization scores higher. Outside the theorems’ regime the shortcut mechanism is thus stronger than the starvation story: adaptive preconditioning renormalizes the small spectral gradients (Section 4.5) and the spectral module chases the residual left by the faster spatial module, degrading rather than refining its features. The nonlinear joint-MLP arm shows the same phenomenology, so the effect is not an artifact of the linear parameterization.

SGD dissociation: the theorem’s regime behaves as proved. Under plain SGD with zero weight decay — the setting of Theorem 27 — the story reverts to the theorem’s: joint training and frozen-random become statistically indistinguishable (0.538 ± 0.075 vs. 0.510 ± 0.051 ; gap $+0.028$, within seed noise), which is precisely the theorem’s functional-equivalence prediction, and the spectral share of total squared gradient flow drops from ≈ 0.94 (AdamW joint) to ≈ 0.27 . The optimizer is the dissociating variable: the flow theorems describe SGD-like dynamics correctly, and adaptive preconditioning is what carries the dynamics out of their scope — consistent with the preconditioned-spectrum discussion in Section 4.5.

Envelope curvature grows with width. The fitted residual-envelope rate for the joint-linear arm grows from $\hat{\mu} = 6.0 \times 10^{-4}$ ($M=48$) to 1.9×10^{-3} ($M=192$), a factor 3.2 for a $4\times$

width increase — qualitatively consistent with the width-growing NTK floor of Proposition 31, which is the theory’s proved mechanism for why the spatial module’s fast fit accelerates with capacity.

The scalar EGR diagnostic is blind here. The scalar effective-gradient-ratio *rises* during joint AdamW training (epoch-1 median ≈ 0.8 to ≈ 4 by epoch 60) even as the jointly trained module underperforms its own frozen initialization. A single global ratio cannot see direction-wise starvation (cf. Remark 35): the spectral gradients are large along the mean-spectrum direction and starved along the discriminative ones, exactly as the initialization geometry of Section 8.3 predicts. Direction-resolved diagnostics are future work.

8.5 What the theorems do and do not predict here

Honest scoping of the quantitative statements. The containment assumption of Theorem 27 is violated on real data: the spatial module’s input-Jacobian operator norm grows by orders of magnitude over training ($3.2 \rightarrow 157$ on average for joint-linear, up to 414; $1.7 \rightarrow 1.2 \times 10^4$ for joint-MLP). Instantiating the theorem’s displacement bound with measured constants accordingly yields a bound 2×10^4 – 2×10^5 times the measured displacement: on this pipeline the bound is true but vacuous, and we report it as such. The synthetic setting of Section 7, where containment holds by construction, is where the bound is quantitatively informative; the real-data value of the theory lies in its verified ingredients (width-independent spectral cap, width-growing spatial curvature, the class-prior NTK floor) and in the regime-resolved dynamics above — the functional equivalence it predicts appears exactly where its assumptions apply, and the failure outside that regime is worse for joint training, not better.

The companion study [Dziuba et al., 2026] additionally provides a pixel-level tokenization benchmark across three datasets and the spatial-transfer experiment on breast QCL from which Table 3 is drawn; those results characterize *what* performs well, while the present analysis addresses *why* end-to-end training of the spectral stage underperforms.

9 Discussion

9.1 Implications for practice

The theorems and experiments establish a concrete prescription. In any compositional spectral-spatial pipeline with substantial capacity asymmetry $C_g \gg C_f$, end-to-end joint training is expected to fail in a specific and predictable way: the spectral module stays near initialization while the spatial module fits the data using shortcuts in the unstructured feature map. Three operational responses follow.

Freeze the spectral module. The most direct response to Theorem 27 is to remove θ from the optimization. Without a spectral update, the timescale separation is eliminated by construction, and g_ϕ optimizes a standard supervised learning problem on fixed input features. If a pretrained spectral encoder is available (from unsupervised pretraining, self-supervised pretraining, or pre-task pretraining on the same data), it should be used frozen.

Use a fixed baseline. When pretraining is unavailable, a fixed but informative baseline $h(X)$ (e.g. PCA, fixed wavelet basis, random projection) can be injected as an additive component $Z = f_\theta(X) + h(X)$. The fixed component ensures the spatial model receives a stable input even if f_θ drifts; the additive structure allows fine-grained refinement of f_θ without exposing the spatial model to the moving-target dynamics that produce the shortcut. We have not empirically tested this prescription in this paper; it follows directly from the theory and is a natural focus for follow-up work.